

Machine Learning Practical #2

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Subject/Course:	Machine Learning Practical		
Topic	Regression Analysis		

Implementing linear regression (e.g. to predict housing prices based on features such as size and location).

```
import pandas as pd

from sklearn.linear_model import LinearRegression

import numpy as np

import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split

from sklearn.metrics import mean_absolute_error,
mean_squared_error, r2_score

df=pd.read_csv('C:\\Users\\admin\\Downloads\\mumbai_
_housing_data.csv')

df.head()

df.info()

df.describe()

df.isnull().sum()

if 'Age_yrs' in df.columns:

    df['Age_yrs']=df['Age_yrs'].fillna(df['Age_yrs'].median())
```

```
[5]: df.describe()
```

	Area_sqft	BHK	Age_yrs	Floor	Total_Floors	Dist_Rail_km	Sea_View	Parking	Price_INR
count	1000.000000	1000.000000	1000.000000	1000.000000	1000.000000	1000.000000	1000.000000	1000.000000	1.000000e+03
mean	918.867930	2.344000	19.645000	9.192000	17.026000	4.102520	0.137000	0.691000	1.932822e+07
std	251.998774	0.796428	11.363895	6.515796	7.027995	2.287982	0.344019	0.462312	6.636654e+06
min	350.000000	1.000000	0.000000	1.000000	5.000000	0.100000	0.000000	0.000000	6.065868e+06
25%	737.792500	2.000000	10.000000	4.000000	11.000000	2.140000	0.000000	0.000000	1.444118e+07
50%	887.370000	2.000000	20.000000	8.000000	17.000000	4.045000	0.000000	1.000000	1.838418e+07
75%	1082.047500	3.000000	29.000000	13.000000	23.000000	6.142500	0.000000	1.000000	2.329743e+07
max	1910.770000	4.000000	39.000000	29.000000	29.000000	8.000000	1.000000	1.000000	5.153944e+07

```
[3]:
```

	Locality	Area_sqft	BHK	Age_yrs	Floor	Total_Floors	Dist_Rail_km	Sea_View	Parking	Price_INR
0	Dadar	583.22	1	10	8	15	2.74	0	1	18045592.0
1	Bandra	784.96	1	20	4	5	4.93	0	0	23882179.0
2	Ghatkopar	793.64	3	14	19	19	3.12	1	1	24259453.0
3	Thane	968.72	1	24	18	18	3.14	0	0	18824996.0
4	Andheri	1089.77	4	34	17	18	2.14	0	1	26278509.0

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1000 entries, 0 to 999
Data columns (total 10 columns):
#   Column          Non-Null Count  Dtype
---  -
0   Locality         1000 non-null   object
1   Area_sqft        1000 non-null   float64
2   BHK              1000 non-null   int64
3   Age_yrs          1000 non-null   int64
4   Floor            1000 non-null   int64
5   Total_Floors     1000 non-null   int64
6   Dist_Rail_km     1000 non-null   float64
7   Sea_View         1000 non-null   int64
8   Parking          1000 non-null   int64
9   Price_INR        1000 non-null   float64
dtypes: float64(3), int64(6), object(1)
memory usage: 78.3+ KB
```

```
num_cols=["Area_sqft","BHK","Age_yrs","Floors","Total_Floors","Dist_Rail_km","Sea_View","Parking","Price_INR"]
```

```
for col in num_cols:
```

```
    if col in df.columns:
```

```
        df[col]=df[col].fillna(df[col].median())
```

```
lower,upper=df['Price_INR'].quantile([0.01,0.99])
```

```
df['Price_INR']=df['Price_INR'].clip(lower,upper)
```

```
plt.figure(figsize=(7,4))
```

```
plt.hist(df['Price_INR']/1e7,bins=30,edgecolor="black")
```

```
plt.xlabel("Price(Crores)")
```

```
plt.ylabel("Count")
```

```
plt.title("Distribtuion of Apartment Prices (Mumbai)")
```

```
plt.tight_layout()
```

```
plt.show()
```

```
plt.figure(figsize=(7,4))
```

```
plt.hist(df['Area_sqft'],bins=30,edgecolor="black")
```

```
plt.xlabel("Area_sqft")
```

```
plt.ylabel("Count")
```

```
plt.title("Distribtuion of Apartment Area")
```

```
plt.tight_layout()
```

```
plt.show()
```

```
if "Locality" in df.columns:
```

```
    plt.figure(figsize=(7,4))
```

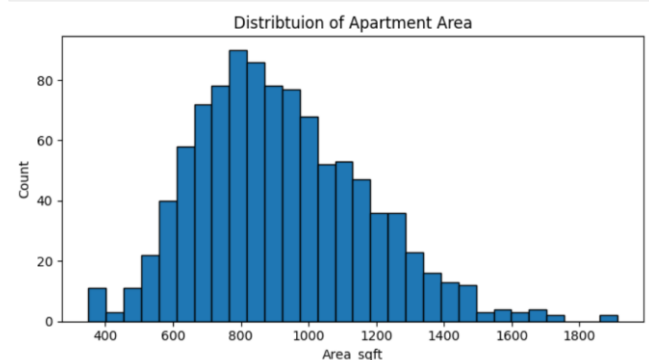
```
    localities=df['Locality'].unique().tolist()
```

```
[6]: df.isnull().sum()
```

```
[6]: Locality      0
     Area_sqft    0
     BHK          0
     Age_yrs      0
     Floor        0
     Total_Floors 0
     Dist_Rail_km 0
     Sea_View     0
     Parking      0
     Price_INR    0
     dtype: int64
```



```
plt.title("Distribtuion of Apartment Area")
plt.tight_layout()
plt.show()
```



```
price_by_loc=[df[df["Locality"]==loc]["Price_INR"]/1e7
for loc in localities]
```

```
plt.boxplot(price_by_loc,labels=localities)
```

```
plt.xlabel("Price (Crores)")
```

```
plt.title("Distribution by locality")
```

```
plt.tight_layout()
```

```
plt.show()
```

```
plt.figure(figsize=(7,5))
```

```
plt.scatter(df['Area_sqft'],df['Price_INR']/1e7,alpha=0.5)
```

```
plt.xlabel("Area (sqft.)") plt.ylabel("Price (Crores)")
```

```
plt.title("Area vs. Price") plt.tight_layout() plt.show()
```

```
df['Price_INR']=df['Price_INR']/1e7
```

```
if "Locality" in df.columns:
```

```
df_encoded=pd.get_dummies(df,columns=['Locality'],dro
p_first=True)
```

```
target=df_encoded['Price_INR']
```

```
feature_cols=df_encoded.drop('Price_INR',axis=1)
```

```
X_train, X_test, y_train, y_test =
```

```
train_test_split(feature_cols,target,test_size = 0.2,
random_state = 42)
```

```
model=LinearRegression()
```

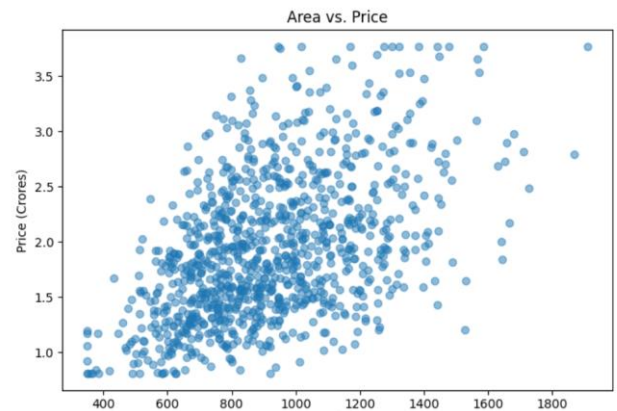
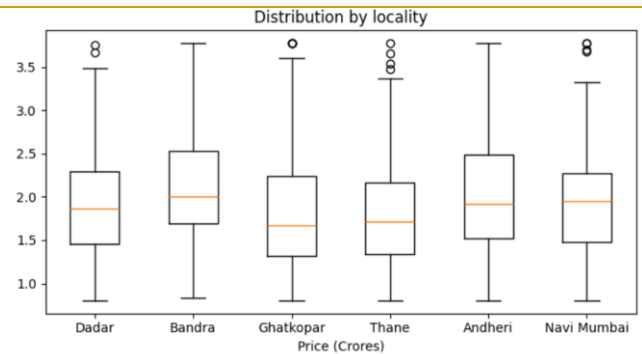
```
model.fit(X_train,y_train)
```

```
y_pred = model.predict(X_test)
```

```
mae = mean_absolute_error(y_test,y_pred)
```

```
rmse = np.sqrt(mean_squared_error(y_test,y_pred))
```

```
r2 = r2_score(y_test,y_pred)
```



```
df_encoded=pd.get_dummies(df,columns=['Locality'],drop_first=True)
```

	Area_sqft	BHK	Age_yrs	Floor	Total_Floors	Dist_Rail_km	Sea_View	Parking	Price_INR	Locality_Bandra	Locality_Dadar	Locality_Ghatkopar	Locality_Navi Mumbai	Locality_Thane
0	583.22	1	10	8	15	2.74	0	1	1.804539	False	True	False	False	False
1	784.96	1	20	4	5	4.93	0	0	2.388218	True	False	False	True	False
2	793.64	3	14	19	19	3.12	1	1	2.425945	False	False	True	False	False
3	968.72	1	24	18	18	3.14	0	0	1.882500	False	False	False	False	False
4	1089.77	4	34	17	18	2.14	0	1	2.627851	False	False	False	False	False
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
995	1092.72	4	0	18	19	6.99	0	0	2.399387	False	False	False	False	False
996	1433.21	2	29	3	14	1.85	0	1	1.645885	False	False	False	False	False
997	797.41	2	21	8	28	5.31	0	0	1.492704	False	False	False	False	False
998	902.27	3	20	8	14	1.58	1	1	1.751122	False	False	False	False	False
999	1086.01	3	14	4	22	4.00	0	1	1.444390	False	False	False	False	False

```
[23]: df_encoded.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1000 entries, 0 to 999
Data columns (total 14 columns):
#   Column              Non-Null Count  Dtype
---  -
0   Area_sqft            1000 non-null   float64
1   BHK                  1000 non-null   int64
2   Age_yrs              1000 non-null   int64
3   Floor                1000 non-null   int64
4   Total_Floors         1000 non-null   int64
5   Dist_Rail_km         1000 non-null   float64
6   Sea_View             1000 non-null   int64
7   Parking              1000 non-null   int64
8   Price_INR            1000 non-null   float64
9   Locality_Bandra      1000 non-null   bool
10  Locality_Dadar       1000 non-null   bool
11  Locality_Ghatkopar   1000 non-null   bool
12  Locality_Navi Mumbai 1000 non-null   bool
13  Locality_Thane       1000 non-null   bool
dtypes: bool(5), float64(3), int64(6)
memory usage: 75.3 KB
```

Evaluation Metrics (Test Set)

Mean Absolute Error (MAE):0.14 Lakh

Root Mean Squared Error (RMSE):0.17 Lakh

R^2 Score:0.9150

```

print("\nEvaluation Metrics (Test Set)")

print(f"Mean Absolute Error (MAE):{mae:.2f} Lakh")

print(f"Root Mean Squared Error (RMSE):{rmse:.2f} Lakh")

print(f"R^2 Score:{r2:.4f}")

plt.figure(figsize=(6,6))

plt.scatter(y_test,y_pred, alpha=0.5)

plt.xlabel("Actual Price (Lakh)")

plt.ylabel("Predicted Price (Lakh)")

plt.title("Actual vs Predicted Price (Test Set)")

min_val = min(y_test.min(), y_pred.min())

max_val = max(y_test.max(), y_pred.max())

plt.plot([min_val,max_val],[min_val,max_val], "r--")

plt.tight_layout()

plt.show()

```

